Name: Mahrukh Haseeb

Introduction:

The objective of this assignment is to curate a high-quality bioactivity dataset from the ChEMBL database to support future Quantitative Structure-Activity Relationship (QSAR) modeling. The selected target is the Androgen Receptor (CHEMBL1871). This nuclear receptor is a critical drug target in oncology, particularly for prostate cancer, as it mediates the effects of androgens like testosterone.

Target and Data Selection:

The Androgen Receptor was selected due to its clinical significance and the abundance of high-quality bioactivity data available in ChEMBL.

Selected Target Name: Androgen Receptor

ChEMBL ID: CHEMBL1871

Number of Raw Bioactivity Records: 4425

Data Curation Workflow:

The data curation process involved following steps:

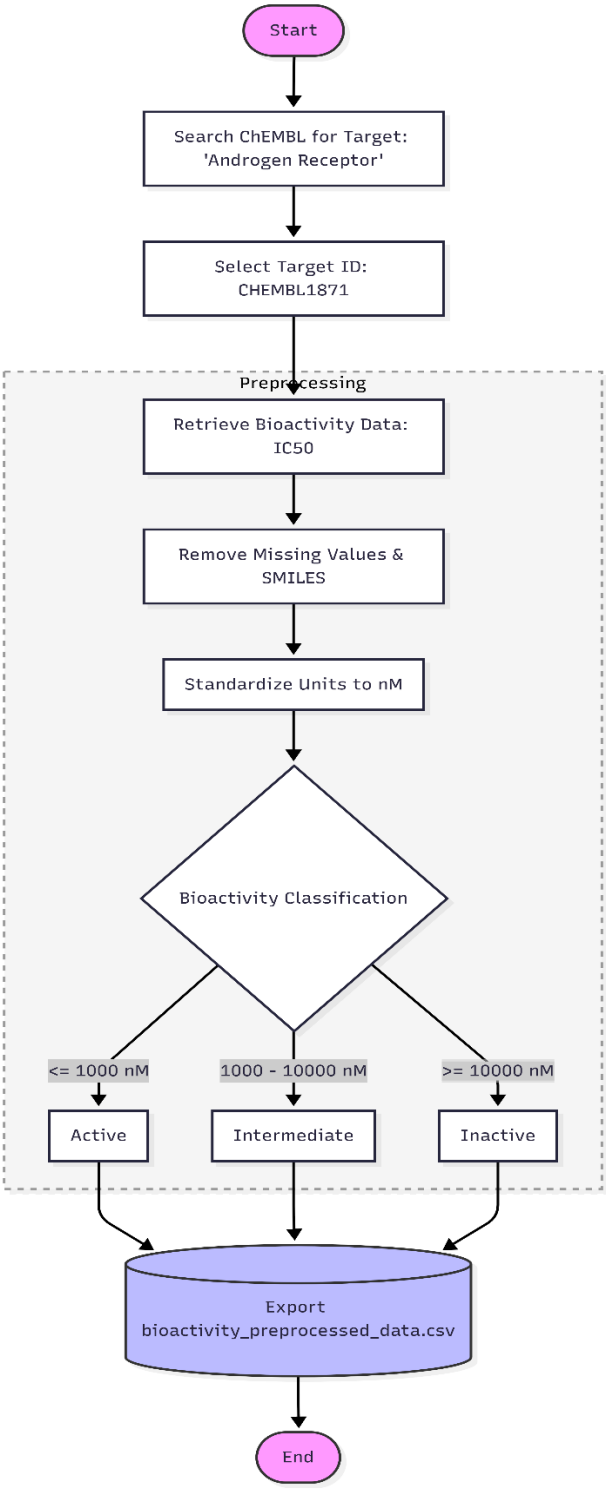
Retrieval:

Bioactivity data for the target was filtered for the "IC50" standard type to ensure comparability.

Preprocessing:

- Filtered out entries with missing standard_value or canonical_smiles.
- Standardized units to Nanomolar (nM).
- Classified compounds into three bioactivity classes: Active (≤1,000 nM), Intermediate (1,000–10,000 nM), and Inactive (≥10,000 nM).

Workflow Overview:



GitHub Repository Link:

https://github.com/Mahrukh-Haseeb/AI_and_Drug_Discovery_Course_2026.git